Asymptotic Notations

Properties & Problems

Asymptotic Notations:Properties

- *Reflexivity:* $f(n)$ is $O(f(n))$.
- *Constants:* If $f(n)$ is $O(g(n))$ and $c > 0$, then $c \cdot f(n)$ is $O(g(n)))$.
- *Products:* If $f_1(n)$ is $O(g_1(n))$ and $f_2(n)$ is $O(g_2(n)))$, then $f_1(n) \cdot f_2(n)$ is $O(g_1(n) \cdot g_2(n)))$.
- *Sums:* If $f_1(n)$ is $O(g_1(n))$ and $f_2(n)$ is $O(g_2(n)))$, then $f_1(n) + f_2(n)$ is $O(\max\{g_1(n), g_2(n)\})$.

Asymptotic Notations:Properties

- *Transitivity:* If $f(n)$ is $O(g(n))$ and $g(n)$ is $O(h(n))$, then $f(n)$ is $O(h(n))$.
- *Polynomials:* Let $f(n) = a_0 + a_1 n + \ldots + a_d n^d$ with $a_d > 0$. Then, $f(n)$ is $\Theta(n^d)$.
- *Logarithms and polynomials:* $\log_b n$ is $O(n^d)$ for every $b > 0$ and every $d > 0$.
- *Exponentials and polynomials:* n^d is $O(r^n)$ for every $r > 0$ and every $d > 0$.
- *Factorials:* $n!$ is $2^{\Theta(n \log n)}$.

Asymptotic Notations:Properties

- *Limits:* If $\lim_{n \to \infty} \frac{f(n)}{g(n)} = c$ for some constant $0 < c < \infty$, then $f(n)$ is $\Theta(g(n))$
- *Limits:* If $\lim_{n \to \infty} \frac{f(n)}{g(n)} = 0$, then $f(n)$ is $O(g(n))$ but not $\Theta(g(n))$.
- *Limits:* If $\lim_{n \to \infty} \frac{f(n)}{g(n)} = \infty$, then $f(n)$ is $\Omega(g(n))$ but not $\Theta(g(n))$.

R-3.2 The number of operations executed by algorithms A and B is $8n\log n$ and $2n^2$, respectively. Determine n_0 such that A is better than B for $n \geq n_0$.

R-3.3 The number of operations executed by algorithms A and B is $40n^2$ and $2n^3$, respectively. Determine n_0 such that A is better than B for $n \geq n_0$.

R-3.7 Show that the following two statements are equivalent:

(a) The running time of algorithm A is always $O(f(n))$.

(b) In the worst case, the running time of algorithm A is $O(f(n))$.

R-3.8 Order the following functions by asymptotic growth rate.

$$\begin{array}{ccc} 4n\log n + 2n & 2^{10} & 2^{\log n} \\ 3n + 100\log n & 4n & 2^n \\ n^2 + 10n & n^3 & n\log n \end{array}$$

R-3.9 Show that if $d(n)$ is $O(f(n))$, then $ad(n)$ is $O(f(n))$, for any constant $a > 0$.

R-3.10 Show that if $d(n)$ is $O(f(n))$ and $e(n)$ is $O(g(n))$, then the product $d(n)e(n)$ is $O(f(n)g(n))$.

Relative asymptotic growths

FUNCTION	$o(n^2)$	$O(n^2)$	$\Theta(n^2)$	$\Omega(n^2)$	$\omega(n^2)$	$\sim 2n^2$	$\sim 4n^2$
$\log_2 n$	✔	✔					
$10n + 45$	✔	✔					
$2n^2 + 45n + 12$		✔	✔	✔		✔	
$4n^2 - 2\sqrt{n}$		✔	✔	✔			✔
$3n^3$				✔	✔		
2^n				✔	✔		

Relative asymptotic growths

Indicate, for each pair of expressions (A, B) in the table below whether A is O, o, Ω, ω, or Θ of B. Assume that $k \geq 1$, $\epsilon > 0$, and $c > 1$ are constants. Write your answer in the form of the table with "yes" or "no" written in each box.

	A	B	O	o	Ω	ω	Θ
a.	$\lg^k n$	n^ϵ					
b.	n^k	c^n					
c.	$\sqrt{n}$	$n^{\sin n}$					
d.	2^n	$2^{n/2}$					
e.	$n^{\lg c}$	$c^{\lg n}$					
f.	$\lg(n!)$	$\lg(n^n)$					

References

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Few Images from the internet